PlacedOn — The Two-Card Verification System (CTO Engineering Plan)

July 16, 2026

CONTENTS

PlacedOn — The Two-Card Verification System (CTO Engineering Plan)

Builds on the research in `placedon_ai_report_card_plan.pdf` (O*NET/ESCO, validation science, fraud data). That doc answers *why* and *what standard*. This one is *how* — how it plugs into the system we already built (Slices 1–8), where I **sharpen** the research, and the order I'd build it as your CTO.

---

0. The one idea (unchanged from your notebook — it's the right one)

The product is **not** two cards. It's the **measured, cited gap between Claim and Proof, scored against a public standard.** We replace the resume with a claim-vs-proof *verification*. Everyone else ships a report; we ship the delta.

The wedge the research now proves: 81–85% of employers say they do "skills-based hiring" and 94% say skills predict performance better than resumes — but almost all still rely on **self-reported** skills AI can now fake (64% of workers admit misrepresenting skills; 11% specifically lie about AI skills; Gartner projects 1-in-4 candidate profiles fraudulent by 2028). The market already believes in verifying skills; the verification infrastructure doesn't exist. Card B *is* that infrastructure, and **Verification %** measures the fake-able gap. (Sources in §11.)

Three artifacts, named precisely (naming matters legally — see §3):

	Name	Author	”What it is”	Status in our build
Card A	**Claim Card**	Candidate	”What I say I can do” — editable self-report	**New** (build it)
Card B	**Evidence Card**	AI, from the interview	”What I demonstrated” — quoted, banded, candidate-reviewed	**Mostly built** (`report_cards`)
—	**Verification Report**	Our software (deterministic)	”How much of the claim is proven, against what the role needs”	**New** (the engine)

---

1. Stand on O*NET + ESCO — do NOT invent requirements

The single most important decision, and I fully agree with the research:

- **O*NET** (US Dept. of Labor) — 1,016 occupations → required skills/knowledge/abilities. Free, machine-readable.
- **ESCO** (EU) — 13,939 skills, finer-grained; what LLM skill-matching research uses.
- **WEF Global Skills Taxonomy (2025)** — explicitly designed as a *”universal adapter”* between region/industry taxonomies; built from 1,000+ employers across 55 economies. Use it as the **crosswalk** so O*NET, ESCO, and an employer’s own language reconcile.

Every role requirement carries a **provenance** tag: `onet:2.B.3.a`, `esco:S1.2.1`, `wef:…`, or `employer` (the Role DNA signal from Slice 1). When a compliance officer asks “where do these come from?”, the card answers **”US Dept. of Labor + the EU standard + the World Economic Forum taxonomy, plus this employer’s stated Role DNA.”** That sentence is the fundability.

Why this is also our legal spine — content validity. Under the EEOC *Uniform Guidelines* (UGESP, 29 CFR 1607), a selection tool that **samples the actual content of the job** is *content-valid by construction*. Because every card item maps to a taxonomydefined job requirement and cites the candidate's own demonstration of it, the card has content validity **on day one** — we don't have to wait for a multi-year criterion study to make a defensible validity claim (though we run one too, §7).

---

2. Where I SHARPEN the research (real CTO judgment, not echo)

Three changes I'd make to the plan you were handed — each for a concrete legal/fairness reason:

1. **Rename "Role-Fit %" → "Role Coverage %".** "Fit" implies a hiring verdict; a verdict that ranks/screens people is an **Automated Employment Decision Tool** under NYC LL144 and high-risk under the EU AI Act. "Coverage" is a *transparency* metric — *which role requirements have supporting evidence* — that informs a human. Same number, defensible framing, and it never auto-rejects.

2. **Rename "Integrity %" → "Consistency %".** Never attach a number to a person's *integrity/honesty* — that's a defamation and fairness landmine. What we can measure honestly is *claim↔evidence alignment* and *within-interview consistency*. A low number means "claims outran the evidence," not "this person is dishonest."

3. **Make Verification % directional and fair.** Proving something you *didn't* claim (proven-unclaimed = under-selling) must **never** lower the score — it's a positive signal. Only *claimed-but-unproven* counts against verification. This protects humble candidates and is the ethical spine of the metric.

The ban still holds: there is **no single blended "hireability" score**, ever. Three separate, cited metrics — never one number that invites "how'd you weight it?"

---

3. The three metrics (each 100% traceable to a quote)

Computed **deterministically** from the claim×evidence×requirement matrix. No number appears without the evidence it stands on.

Metric	Definition	Guards
Verification %	proven claims ÷ claims the candidate made	under-claiming never penalized; each proven claim cites a `report_card_item`
Role Coverage %	role requirements with supporting evidence ÷ total role requirements	requirements come from O*NET/ESCO + employer; must-haves shown separately; informs, never decides
Consistency %	claim↔evidence alignment + within-interview consistency	measures alignment, not a person's honesty; low = "claims outran proof"

For every requirement, the engine assigns one of four states — this matrix *is* the product:

```
                          proven in interview?
                          YES                   NO
claimed?  YES   ✅  Verified            ⚠️  Claimed, not shown
          NO    ➕  Proven (bonus)      ⬜  Gap (evidence to build)
```

---

4. Data model — on top of what's already built

```
taxonomy_skills        (id, source onet|esco, code, label, description)
-- cache of the standard
role_requirements      (job_id|role_family, skill_id, source, weight,
is_must_have, provenance)
candidate_claims       (candidate_id, skill_id|free_text, self_level,
target_role) -- CARD A (new; RLS candidate-owned)
report_cards / report_card_items
-- CARD B (built)
verification_reports   (id, session_id, candidate_id, job_id,
taxonomy_version,
                        model_run, verification_pct, coverage_pct,
consistency_pct,
                        signature, built_at)
-- the comparison
verification_lines     (report_id, skill_id, state ∈
verified|claimed_unproven|
                        proven_unclaimed|gap, band, claim_id?, item_id?)
-- one row per requirement, each citing evidence
```

- Card A and the verification report are **candidate-owned (RLS)**; employers read a **visibility-scoped view** (same pattern as Slice 5: approved + not hidden/disputed).
- `signature` = hash of (taxonomy_version + model_run + cited item ids). Tamper-evident → this is the literal "trust verification from our company" — a reproducible artifact.

---

5. What the card must show HR (the "industry requirements")

The trust fields that make an enterprise/HR/bank say yes:

1. **Skill → taxonomy mapping** with source (O*NET/ESCO/employer) per line.
2. **Evidence provenance** — the candidate's own quote + which interview turn.
3. **Band + confidence** per item (supported/emerging/needs-more) — never a raw score.
4. **The three metrics**, each expandable to the evidence behind it.

5. **Fairness statement** — adverse-impact status; “assessed as text; voice never scored for tone.”
6. **Validity claim** — the predictive-validity number from our study (§7), dated.
7. **Reproducibility** — taxonomy + model versions; the signature; “re-runnable.”
8. **Consent & scope** — what the candidate approved for this employer; identity withheld until intro.
9. **Authenticity** — session integrity signals (one live session, no paste-dumps) without scoring behavior.

---

6. Audience views

- **Candidate:** author/edit **Card A**; review **Card B** (built); see the verification matrix as *guidance* — “evidence to build” on the gaps, not a grade.
- **Employer/HR:** Evidence Card + Role Coverage + the verification badge; every number expands to its quote; identity withheld until a consented intro (Slice 7).
- **Verifier (3rd party / compliance):** a **signed, reproducible verification page** — the artifact that replaces “trust us” with “re-run it yourself.”

Same card, two emphases (big-tech vs startup — research-driven). The audiences want different things, so the employer view has two densities of the *same* evidence:

	Enterprise / big-tech HR	Startup / founder
Wants	compliance, defensibility, audit trail	”can they do the job?”, speed, signal
Surfaces	taxonomy provenance, adverse-impact statement, validity evidence, reproducibility signature	top proven capabilities + the gap, one screen, work-sample-like
Default	**Trust panel expanded**	**Signal-first, trust panel collapsed**

This mirrors how they actually hire: big tech runs structured rubric interviews (Google re:Work) and needs the paper trail; startups run lightweight work-sample + behavioral rounds (YC) and need fast, honest signal. One data model, one card, a density toggle.

---

7. Validity — position on the method meta-analysis already ranks #1

The 2022 re-analysis (Sackett, Zhang, Berry & Lievens) corrected decades of range-restriction over-correction and **re-ranked the predictors**. The result is the best thing that could have happened to us:

- **Structured interviews are now the single highest-validity predictor ($r \approx .42$)** — above general cognitive ability (revised down to $r \approx .31$). Résumé/education screening sits near the bottom (education ≈ .10).
- Our AI interview + Role DNA rubric **is a structured, behaviorally-anchored interview run at scale** — the exact method Google's re:Work program adopted (planned questions + scoring guide + rubric) and the method the science ranks first. We don't claim a novel predictor; we **operationalize the best-validated one** and render it as cited evidence.
- The research is also clear that the **most valid processes combine methods** (structured interview + work sample + cognitive). So the card is positioned honestly as **one strong, calibrated component**, not a silver bullet — "use alongside your judgment," which *builds* trust rather than over-claiming.

Two validity claims, sequenced:

1. **Content validity — now.** Built in via taxonomy-mapped evidence (§1). Defensible on day one.
2. **Criterion validity — the study.** Correlate card signals with real outcomes from the **Slice 8 outcome check-ins (built)**; target the structured-assessment band (~.4). Gate everything behind the **four-fifths / 80% adverse-impact rule** (UGESP) with continuous monitoring. This is the fundable, enterprise-closing number — instrument for it from day one.

---

8. Build order (grounded in what exists; ~5 weeks to a working verified card)

Phase	Weeks	Deliverable
P1	2–3	Taxonomy service (O*NET/ESCO ingest + cache) + `role_requirements` for the top 5 roles (junior backend first).
P2	2–3	**Card A** authoring UI + **Card B** rendering from the *existing* interview engine → both cards live from one real interview.
P3	1–2	The **comparison engine** + the three cited metrics + `verification_reports` / `lines` (deterministic, reproducible).
P4	1–2	Audience views + the **signed verification artifact**.
P5	ongoing	Validation-study instrumentation (predictive validity + adverse-impact), feeding off Slice 8.

Start with the **inputs** (taxonomy + a filled Card B), not the comparison — the comparison is trivial once its two inputs are trustworthy.

---

9. Honest dependencies (the floor this stands on)

- The **evidence pipeline** (LLM extract → verify_quote gate → fidelity judge → calibrate) that *fills* Card B is still a seam — needs the **hosted backend + keys**. `verify_quote` and `gate_items` (built) are where it plugs in.
- This sits directly on the **auth/RLS + evidence-verification** foundations we already built. A verification card on unverified evidence or leaky access is **worse than none** — those foundations are not separate work; they're the ground floor.

10. What I would NOT do

- No single blended score. No scraped/inferred data. No scoring voice/accent/behavior.
- No auto-reject / auto-rank. No invented requirements. No shipping the comparison before the taxonomy and a real filled Card B exist.

11. Research foundation & sources (why this is fundable, not invented)

A. Validity science — our method is the #1-ranked predictor.

- Sackett, Zhang, Berry & Lievens (2022) re-analysis: **structured interviews r ≈ .42 (highest single predictor)**; general cognitive ability revised down to r ≈ .31; most prior validities were over-estimated via range-restriction over-correction. Structured interview + cognitive composite ≈ .6.
- Google re:Work: structured interviewing (planned questions + rubric + behaviorally-anchored scales) “consistently outperforms freeform conversations at every level.”
- McKinsey (via skills-hiring reports): skills-based hiring is **5× more predictive than education, 2×+ more than experience**; the most valid processes **combine methods**.

B. The market believes in skills but can’t verify them (our wedge).

- **81–85%** of employers claim skills-based hiring; **94%** say it predicts better than resumes; **72%** use skills assessments — yet the implementation gap is huge (some reports: ~1 in 700 hires actually affected). Most still trust **self-reported** skills.

C. Fraud/trust — the timing (our verification angle).

- **63%** of fraudulent-resume applicants got offers in 2024; **64%** of workers admit misrepresenting skills (up from 55% in 2022); **11%** lie specifically about AI skills.
- **59%** of managers suspect candidates use AI to misrepresent; **62%** say candidates now fake identity better than teams detect. **Gartner: 25% of candidate profiles fraudulent by 2028.** Reported job-fraud losses **$90M → $501M (2020→2024, +457%)**.

D. Standards to stand on.

- **O*NET** (US DoL), **ESCO** (EU, 13,939 skills), **WEF Global Skills Taxonomy 2025** (universal adapter). **EEOC UGESP** (29 CFR 1607): content/criterion/construct validity; **four-fifths (80%) rule** for adverse impact — validation required once adverse impact exists.

Sources:

- Sackett et al. (2022), Revisiting Meta-Analytic Estimates of Validity · SIOP summary
- Google re:Work — Structured Interviewing guide
- TestGorilla — State of Skills-Based Hiring 2025
- Resume.org — 6 in 10 Resume Fraudsters Landed a Job in 2024 · CNBC — deepfake job applicants
- WEF Future of Jobs 2025 — Skills Outlook · WEF Global Skills Taxonomy Adoption Toolkit
- EEOC Uniform Guidelines (29 CFR 1607) · Q&A on UGESP
- O*NET · ESCO